period, between April and July that year, gave him valuable lessons in not just development of propellants, but their transport, filling and storage.

Initially his team did some basic experiments with aniline and furfural alcohol made from rice husk collected from a mill in Nagercoil. Mixed with furfural, aniline caught fire. Then it moved on to bigger fuels like red-fuming nitric acid (RFNA), which we used for smaller rockets.

Vikas needed the more complicated unsymmetrical dimethylhydrazine (UDMH). In those days, the Soviet Union was using RFNA, along with aniline, to fire its missiles. RFNA, which comprised 84% nitric acid 13% nitrogen tetroxide and about 2% water, is a highly poisonous chemical, with a threshold limit value of 5ppm (parts per million). This meant exposure to 5ppm of RFNA can be dangerous, even fatal. UDMH is a carcinogen with a much lesser threshold limit value—2ppm.

Besides trying out different chemical reactions to optimally produce the propellants, the team was mastering the tricky pipeline layouts, tank layouts and mitigation, in case of a leak in a pipeline or a breach in a tank.

For the first ground test of Vikas at Vernon, SEP had provided UDMH and the oxidiser which cost 60% of our Rs 1 crore budget for the test. While we saved some of the fuel and oxidiser we had imported for the first flight, our team developed UDMH and N2O4 (nitrogen tetroxide) for several ground tests in Mahendragiri before the launch. This was a result of great toil, starting from textbooks to trials and errors before success.

For N2O4, they tried treating lead nitrate with lead monoxide (a lethal chemical when exposed in nitric acid). A pilot plant was set up for this at VSSC, Trivandrum. As for UDMH, there were two teams at work. One under Natarajan using the chloramine process, and the other under Gowarikkar using the dimethyl urea process.

In the laboratory, both the processes were found to be successful, and now we wanted to scale up either of the two. Presented with the option, VSSC director Brahm Prakash appointed a selection committee headed by Indian Petrochemical Corporation Limited (IPCL) chairman S Varadarajan. Indian Drugs and Pharmaceuticals Limited (IDPL), Hyderabad, the largest